# vi Cheat-Sheet: Beginner Edition

The ~60 commands that cover 90% of daily use

> Rule 1: vi has modes. You start in Normal mode (keys are commands). Press `i` to type text (Insert mode). Press `<Esc>` to get back. When unsure, press `<Esc>` twice.

> Panic exit: `<Esc>` `<Esc>` then `:q!` and Enter = quit without saving. `<Esc>` then `:wq` and Enter = save and quit.

## 1. Survive First: Open, Save, Quit

| Keys | What it does | Keys | What it does |
|---|---|---|---|
| `vi file` | Open (or create) a file | `:q` | Quit |
| `i` | Start typing (Insert mode) | `:wq` or `ZZ` | Save and quit |
| `<Esc>` | Back to Normal mode | `:q!` | Quit, throw away changes |
| `:w` | Save | `:set nu` | Show line numbers |

## 2. Enter Insert Mode (all end with <Esc>)

| Keys | What it does | Keys | What it does |
|---|---|---|---|
| `i` | Insert before cursor | `A` | Append at end of line |
| `a` | Append after cursor | `o` | New line below |
| `I` | Insert at start of line | `O` | New line above |

## 3. Move Around (Normal mode)

| Keys | What it does | Keys | What it does |
|---|---|---|---|
| `h j k l` | Left, down, up, right | `{n}G` or `:{n}` | Jump to line number n |
| `w / b` | Next word / previous word | `Ctrl-f / Ctrl-b` | Page down / up |
| `0 / $` | Start / end of line | `%` | Jump to matching bracket |
| `gg / G` | Top / bottom of file | `{ / }` | Previous / next paragraph |

## 4. Edit: Delete, Copy, Paste, Undo

| Keys | What it does | Keys | What it does |
|---|---|---|---|
| `x` | Delete character | `cc` | Change whole line |
| `dd` | Delete (cut) line | `r{c}` | Replace one character |
| `5dd` | Delete 5 lines | `J` | Join with next line |
| `dw` | Delete word | `u` | Undo |
| `D` | Delete to end of line | `Ctrl-r` | Redo |
| `yy` | Copy (yank) line | `.` | Repeat last change |
| `p / P` | Paste below / above | `>> / <<` | Indent / outdent line |
| `cw` | Change word | `~` | Toggle upper/lower case |

> Put a number before most commands to repeat them: `3dd` deletes 3 lines, `10j` moves down 10, `2p` pastes twice.

## 5. Search and Replace

| Keys | What it does | Keys | What it does |
|---|---|---|---|
| `/text` | Search forward | `:%s/old/new/g` | Replace all in file |
| `?text` | Search backward | `:%s/old/new/gc` | Replace all, ask each time |
| `n / N` | Next / previous match | `:s/old/new/` | Replace on current line only |
| `*` | Search for word under cursor | `:noh` | Clear search highlighting |

## 6. Select (Visual Mode)

| Keys | What it does | Keys | What it does |
|---|---|---|---|
| `v` | Select characters | `y / d` | Copy / delete the selection |
| `V` | Select whole lines | `> / <` | Indent / outdent selection |
| `Ctrl-v` | Select a column block | `<Esc>` | Cancel selection |

## 7. Everyday Admin Tasks

| Keys | What it does | Keys | What it does |
|---|---|---|---|
| `:r file` | Insert another file here | `:%d` | Delete everything in the file |
| `:r !date` | Insert output of a shell command | `:e!` | Reload file, discard changes |
| `:!ls` | Run a shell command and come back | `:set paste` | Turn on before pasting code (stops staircase indent) |
| `:w !sudo tee %` | Save a file you forgot to open with sudo | `:set nopaste` | Turn off after pasting |

## 8. Five Habits That Save You

- Press `<Esc>` before every command. Most beginner errors come from being in the wrong mode.
- Use `u` freely. Undo is cheap, so experiment.
- Use `dd`, `yy`, `p` for moving lines, not manual retyping.
- Use `:%s/old/new/gc` instead of hunting for text by eye.
- Practise with `vimtutor` in your terminal. It takes 30 minutes and is the best way to build muscle memory.

> Next step: when these feel automatic, move to the full cheat-sheet for text objects (`ci"`, `dap`), macros, `:g` global commands, buffers and splits.